



Motivation

1. Improve upon worst-case total ionizing dose (TID) screening methods for qualification of complex devices using knowledge of susceptibility based on programming choices.
2. In situ, use electrical characteristics to individualize dose-to-failure estimates for a complex devices with large sample-to-sample variability.

Abstract

- Ninety-seven MSP430FR6989 microcontrollers from two fabrication lots were irradiated to functional failure, which was observed from 72 to 193 krad(SiO₂)
- Bias mode and fabrication lot determine the failure point; the most conservative bias states have the least tolerance
- Beyond-worst-case TID screening methods are demonstrated using programmable bias knowledge and lot identification methods
- During irradiation, characteristics of a local oscillator provide individualized dose-to-failure estimates with no added hardware

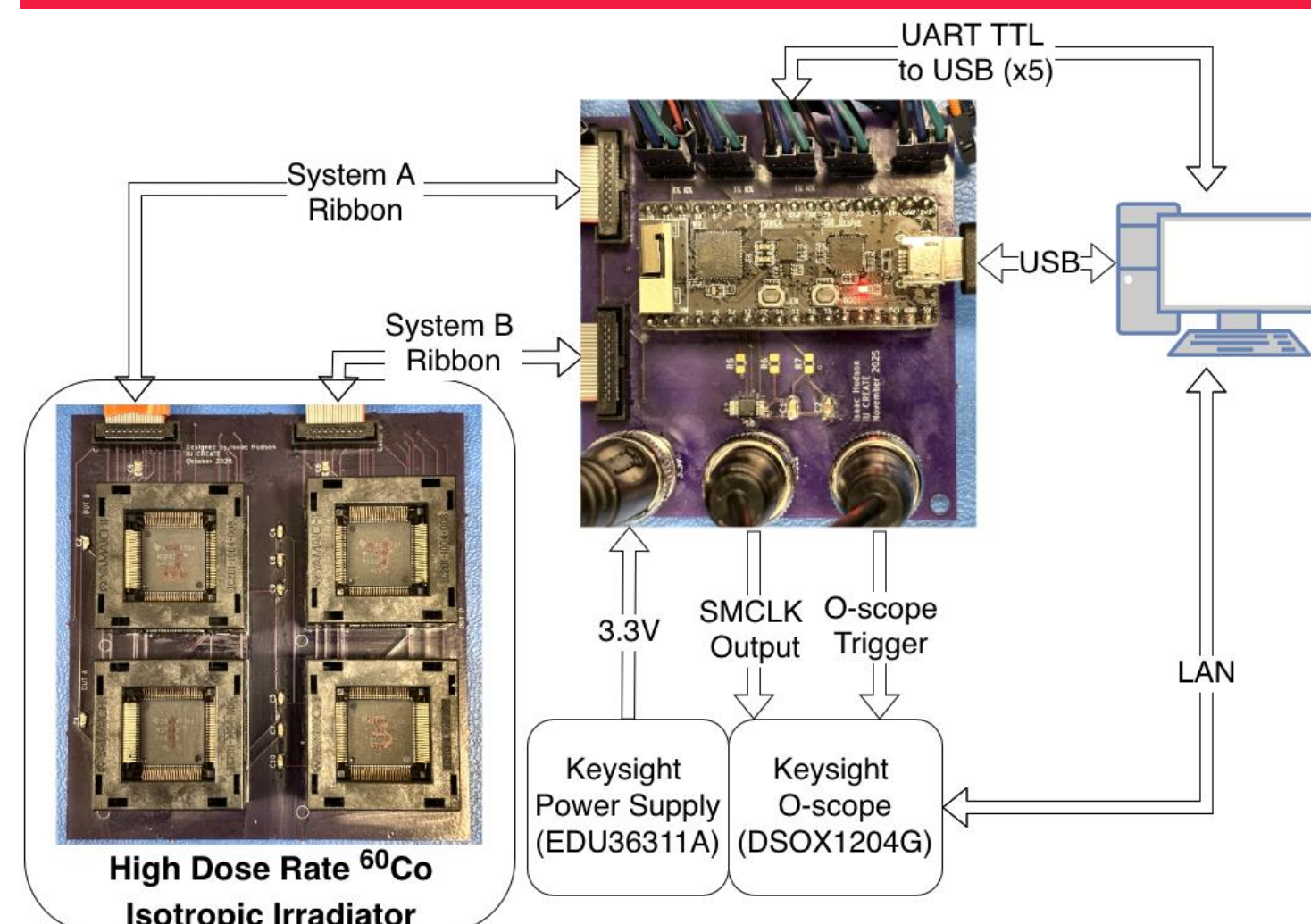


Fig. 1: Experimental Hardware Block Diagram

Test Articles:

- Ninety-seven statically biased MSP430FR6989 microcontroller units (MCUs), an ultra-low-power commercial-off-the-shelf (COTS) part, from two fabrication lots
- Each device held in one of six bias configurations, active (AM) to deepest sleep (LPM4) (Table 1)
- Stepped irradiation to failure, with very low-frequency oscillator (VLO) and digitally-controlled oscillator (DCO) waveforms captured at the SMCLK pin every dose step

Table 1: DUT Bias Mode Description

Mode	AM	LPM0	LPM1	LPM2	LPM3	LPM4
CPU state	on	off	off	off	off	off
FRAM state	on	standby*	off	off	off	off
Max system clock	16 MHz	16 MHz	16 MHz	50 kHz	50 kHz	0
Typical current (25°C)	103 μ A/MHz	75 μ A @ 1 MHz	40 μ A @ 1 MHz	0.9 μ A	0.4 μ A	0.3 μ A
MCLK	on	off	off	off	off	off
SMCLK	optional	optional	optional	off	on	off
ACLK	on	on	on	on	on	off
SVS	always	always	always	optional	optional	optional

*In LPM0, FRAM is typically in standby; some configurations may fully power it off.

3 clusters of 6 bias modes form based on TID-response; the most conservative states have the least tolerance.

- Failure dose spans 72–193 krad(SiO₂), determined by bias mode and manufacturing lot (Fig. 2)
- The lot effect is large and mode-dependent: in AM, lot 2 averaged 180 krad(SiO₂) vs 98 krad(SiO₂) for lot 1
- First failures are missed I/O interrupts; total clock failure follows by a device-dependent margin

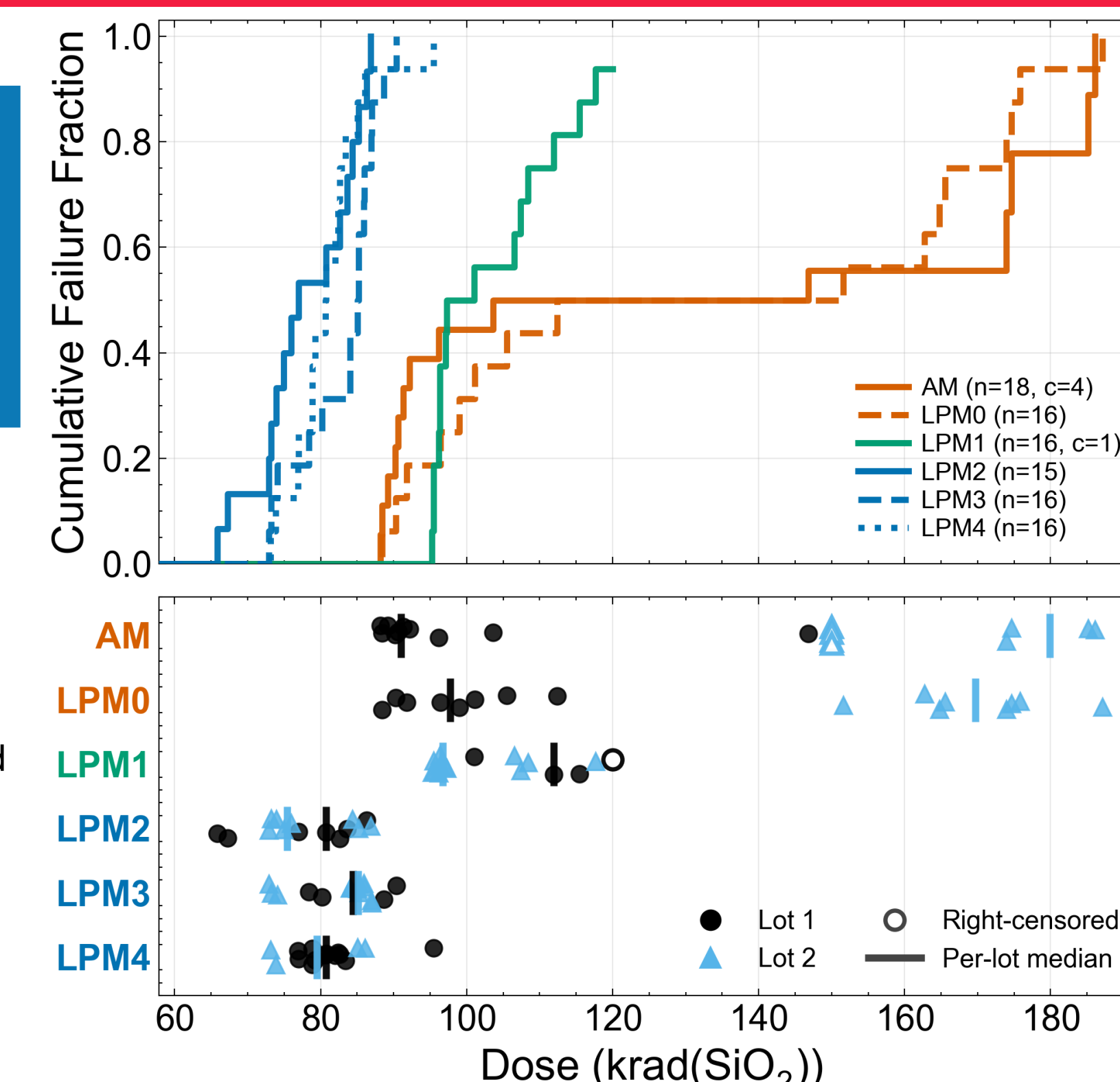


Fig. 2: Failure Point Distribution by Bias Mode and Manufacturing Lot

Feature Extraction

- Each device under test (DUT) outputs its clocks on a pin; an oscilloscope captures the VLO (≈ 10 kHz) and DCO (1 MHz) at every dose step (Fig. 3, A)
- Rising-edge timing gives the period family (mean, jitter, spread) (Fig. 3, B); voltage distribution gives the rail-to-rail swing, p99 – p1 (Fig. 3, C)
- Period features are computable on-chip from existing timers: self-reported period during experiments matched the oscilloscope at $r = 0.999$ (0.55% median deviation)

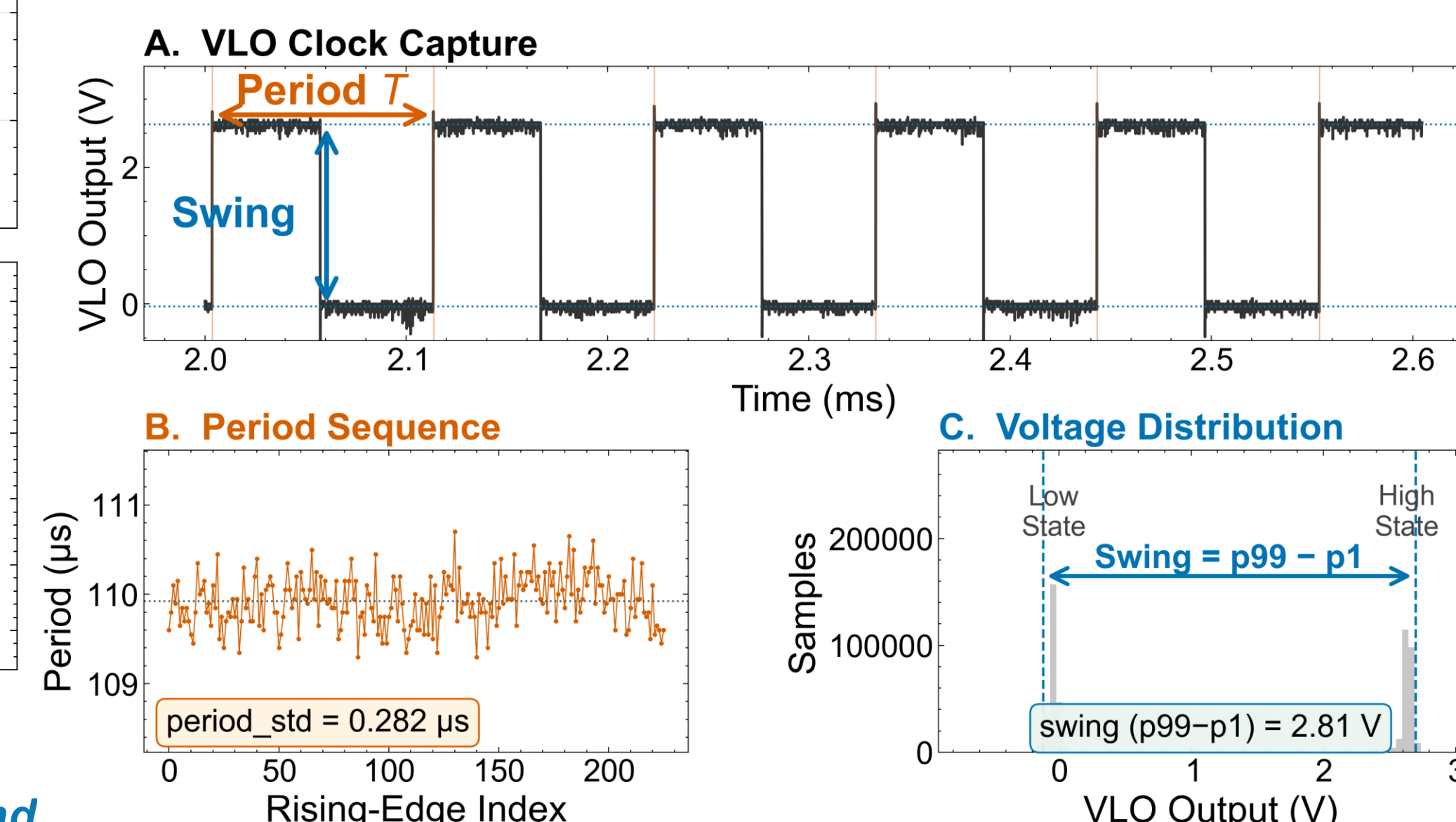


Fig. 3: Feature Extraction on the VLO Waveform

Pre-Irradiation Predictions

A pre-irradiation oscillator measurement allows failure dose predictions beyond naïve worst-case and bias-mode informed methods.

- VLO period jitter acts as a proxy for lot identification using gradient-boosted survival models with bias mode and VLO period jitter features (Fig. 4)
- Models formed per-bias mode sharpen to a mean absolute error (MAE) = 2.2 krad(SiO₂) with the lot known, MAE = 4.8 and C = 0.83 lot-blind

Table 2: Pre-Irradiation Failure Prediction by Configuration

Model	Information used	MAE	C	Inside	≤ 10 krad
Naive worst-case	population constant	30.9	0.50	2%	25%
Mode-only	bias mode	12.6	0.71	35%	62%
Mode + VLO	bias mode + oscillator	4.4	0.81	55%	89%
Lot + Mode*	lot label	2.4	0.80	46%	95%

MAE in krad(SiO₂); C represents Concordance: A measure of how well the model orders devices based on failure risk (higher tolerance devices have lower risk) 1 is a perfect ranking, 0.5 is pure chance; Inside = share of predictions inside the failure window; ≤ 10 krad = share within 10 krad(SiO₂) of it

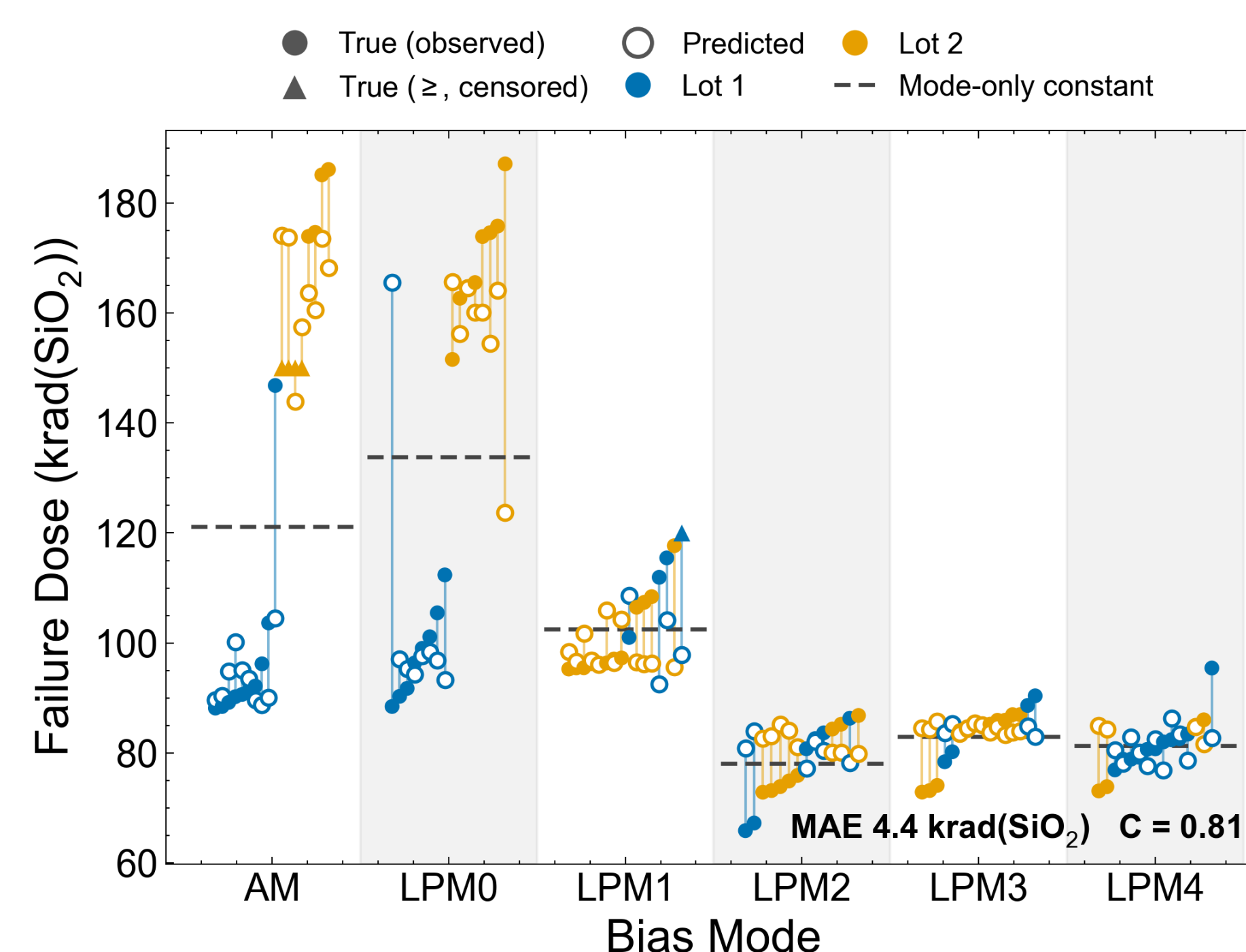


Fig. 4: Pre-Irradiation Model Performance by Bias Mode

Predicted vs observed failure dose, one point per device. Predictions from the pooled lot-blind Mode + VLO model (one model, no per-mode selection).

In-Situ Health Monitoring

In flight, the same oscillator can stand-in for a dosimeter and work as an end-of-life alarm.

- Bias + VLO period alone matches dose-informed accuracy (overlapping confidence intervals) — a self-contained health monitor using gradient-boosted regression trees (MAE loss) (Table 3)
- Swing features of both oscillators improve near-failure predictions (Fig. 5, B), potentially generalizing to any clock output
- Models using VLO features overpredict remaining dose (optimistic) for devices whose clock outlives the observed I/O failure (clock survives in Fig. 5, A)
- The close band in Fig. 5, B and Table 3 represent the last 2 captures (~ 24 krad(SiO₂)) before failure

Table 3: In-Situ Failure Prediction by Configuration

Configuration	MAE (all)	Coverage (all)	MAE (close)	Coverage (close)
Dose	14.8	33%	7.6	38%
Dose/Bias	11.5	43%	5.5	47%
Bias/Period	15.8	27%	14.7	18%
Dose/Bias/Period	11.4	41%	6.1	44%
Dose/Bias/Swing	11.6	43%	5.3	53%

MAE in krad(SiO₂); coverage = share of predictions inside the failure window

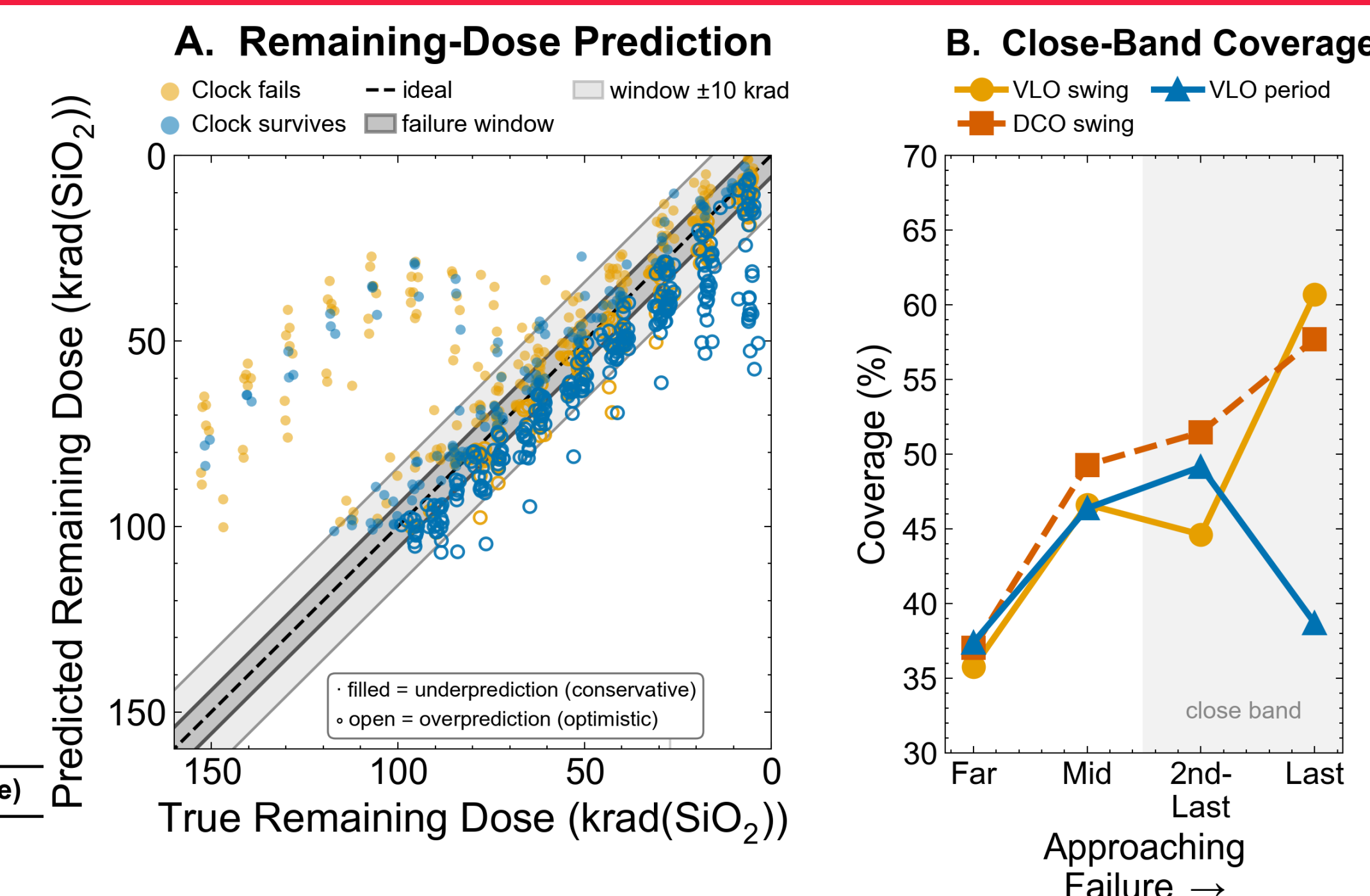


Fig. 5: In-Situ Remaining-Dose Prediction Accuracy and Coverage by Feature Group

(A): Predicted vs true remaining dose at every capture for the Dose/Bias/Swing Config (Table 3); median failure window and ± 10 krad window drawn from the ideal (B): Share of predictions inside the failure window as failure nears

Model Details

- Train: 50 random device-level splits — 80% (pre-rad) and 70% (in-situ) training; the remaining 20% and 30% are held-out
- Validation: Model choices are made one split at a time from each candidate's held-out performance on the other 49 splits (leave-one-split-out)
- Test: The selected model's predictions on each split's held-out devices are scored; reported metrics aggregate these predictions across all 50 splits — no model is ever scored on a device it trained on

Conclusions

- Knowledge of application bias states and lot membership informs TID survival predictions past worst-cast
- Period-derived measurements of the device's VLO inform prognostic models through life
- Rail-to-rail voltage swing at the shared output path of 2 distinct oscillators informs these models in late-life
- The lot recovery demonstrated here separates two extensively screened lots; extending the screen to devices of unknown provenance is future work.